



Principles Course Introduction & Overview David Casaceli, PE

DALLAS, TX



AGENDA

RILEY EDWARDS
UNIV. OF ILLINOIS U-C

DARRELL KRUEGER
BNSF RAILWAY

KEVIN OLDKNOW
SIMON FRASER UNIV.

COREY PASTA
MxV RAIL

RICHAR STOCK
PLASSER & THEURER

MATT DICK
ESi

1 TRACK STRUCTURES &
COMPONENTS

2 VEHICLE TYPES, SUSPENSIONS,
AND COMPONENTS

3 WHEEL-RAIL CONTACT: AN
INTRODUCTION

4 VEHICLE-TRACK INTERACTION &
DYNAMICS

5 WHEEL-RAIL DAMAGE
MECHANISMS

6 VEHICLE-TRACK MEASUREMENT
TECHNOLOGIES



DERAILMENT CASE STUDY

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- 1 TRACK STRUCTURE COMPONENTS
- 2 VEHICLE TYPES, SUBSYSTEMS, AND COMPONENTS
- 3 WHEEL-RAIL CONTACT INTRODUCTION
- 4 VEHICLE-TRACK INTERACTION DYNAMICS
- 5 WHEEL-RAIL DAMAGE MECHANISMS
- 6 VEHICLE-TRACK MITIGATION TECHNOLOGIES

DERAILMENT CASE STUDY



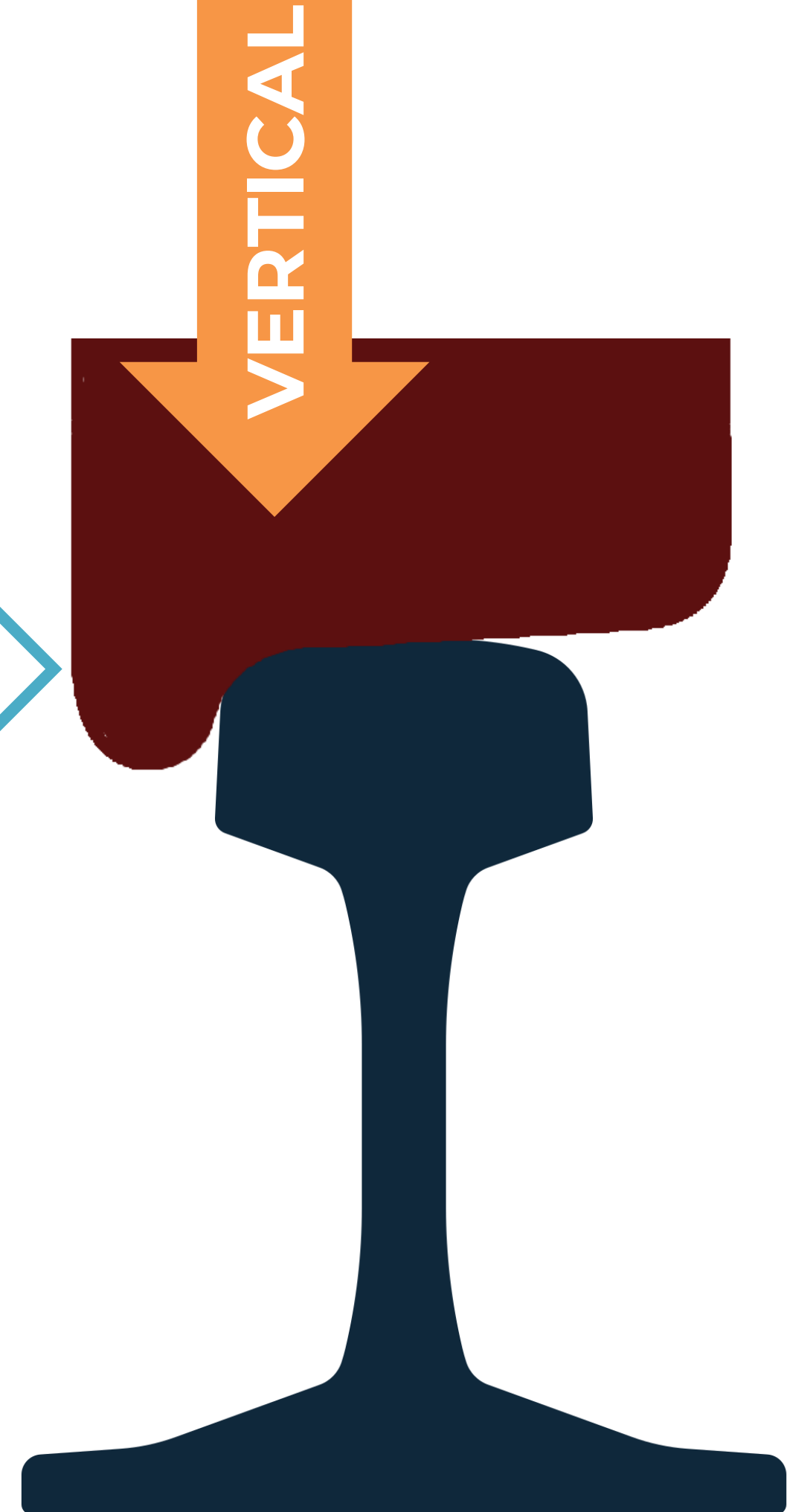
WHY WHEELS DERAIL

WHAT IS L / V?





$$\frac{L}{V} = \frac{\text{LATERAL}}{\text{VERTICAL}}$$





APPLIED L/V

The ratio of the forces actually applied at the wheel-rail interface.

Vertical Forces are a function of:

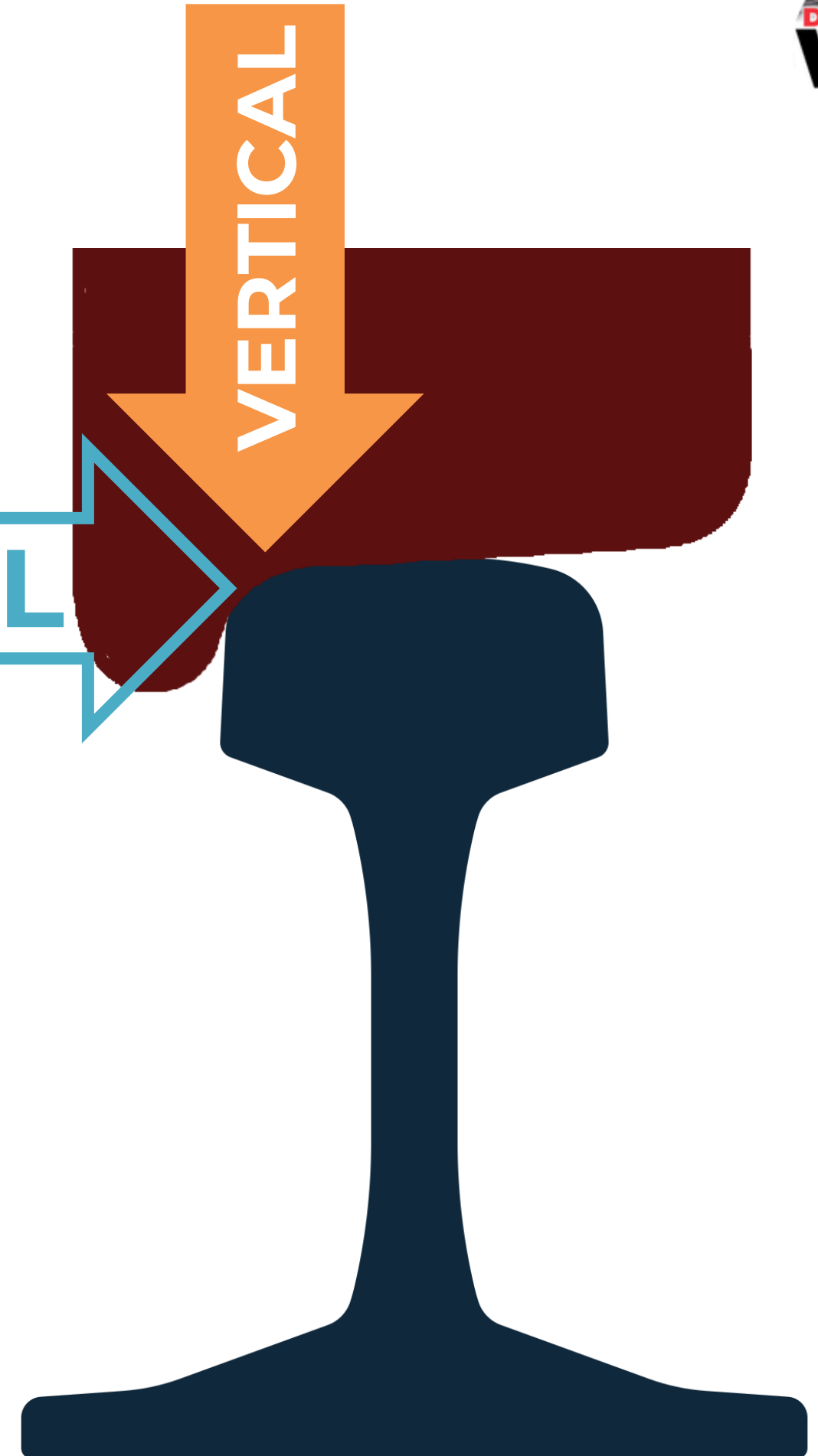
- Vehicle weight
- Track geometry and speed
- Truck suspension
- Car design features - side bearings

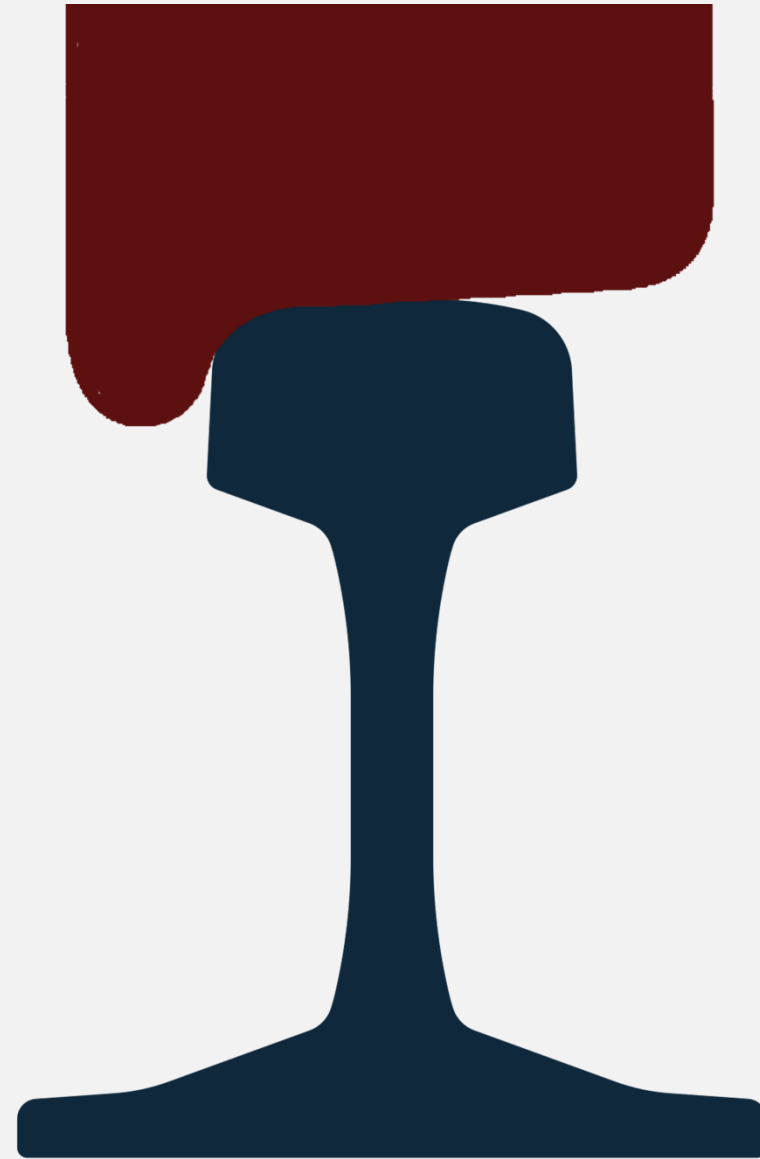
Lateral Forces are a function of:

- Flange and creep forces
- Track geometry and speed
- In-train drawbar forces
- Coupler angles
- Rail lubrication
- Truck/vehicle design and condition

LATERAL

VERTICAL





THE TWO L / V RATIOS



▶ APPLIED L/V

The ratio of the forces actually applied at the wheel-rail interface.

▶ L/V THRESHOLD

The threshold (for a specific wheel-rail combination) that, if exceeded by the APPLIED L/V, will result in wheel climb or rail rollover

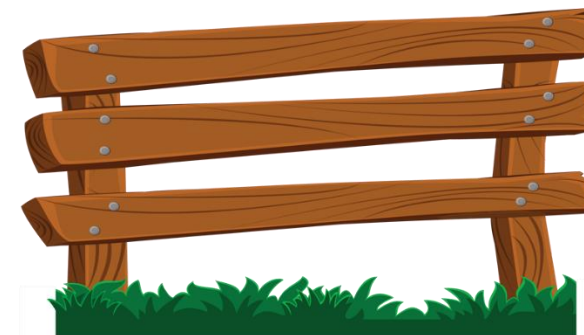
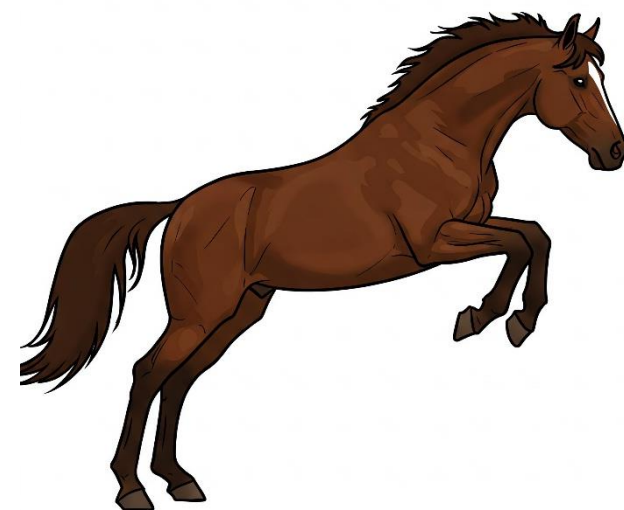
Dependent upon:

- Wheel Profile
- Rail Profile
- Friction at the interface

THE TWO L / V RATIOS



► **APPLIED L/V**
The ratio of the forces actually applied at the wheel-rail interface.



► **L/V THRESHOLD**
The threshold that, if exceeded, will result in a wheel climb or rail roll for a specific wheel-rail combination.

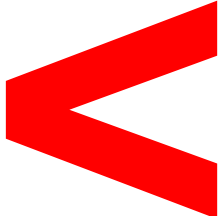
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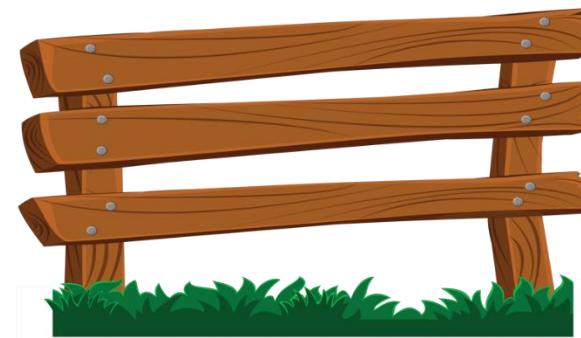
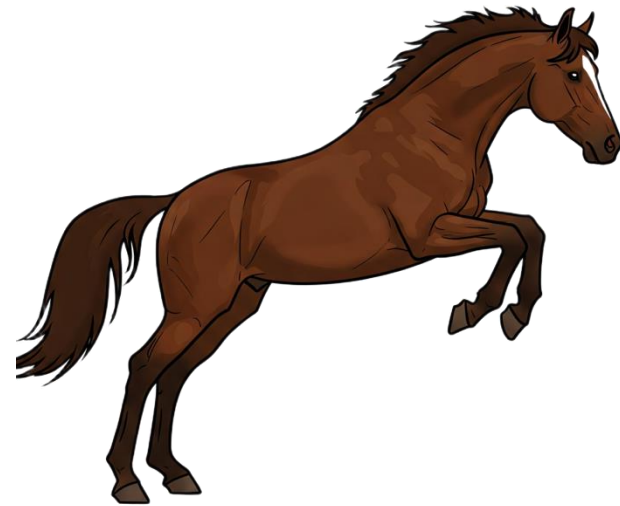


THE TWO L / V RATIOS



HOW DO WHEELS STAY ON THE RAIL?

APPLIED L/V  THRESHOLD L/V



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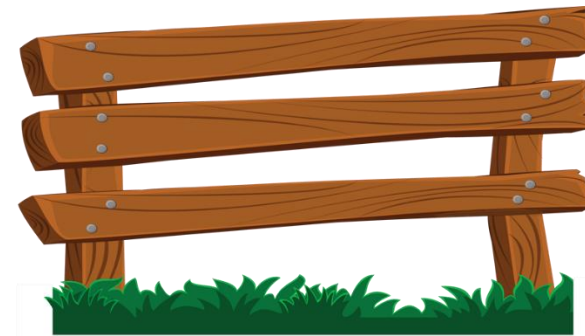
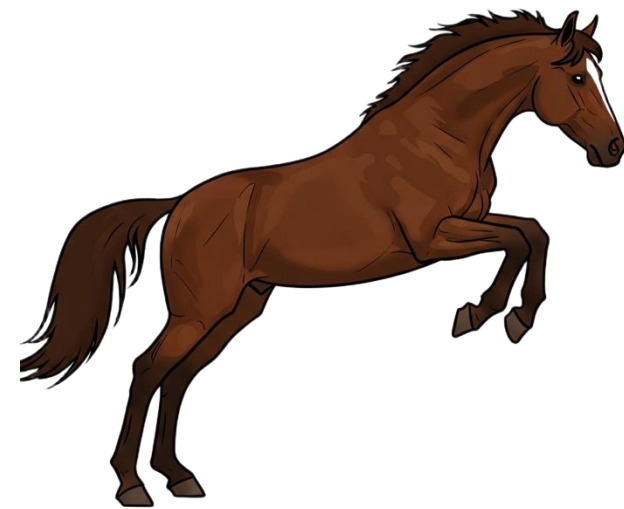


THE TWO L / V RATIOS



WHEN DO DERAILMENTS OCCUR?

APPLIED L/V $>$ THRESHOLD L/V



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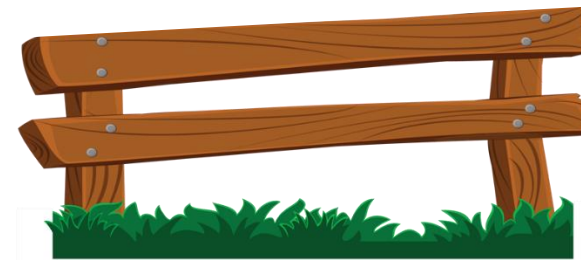
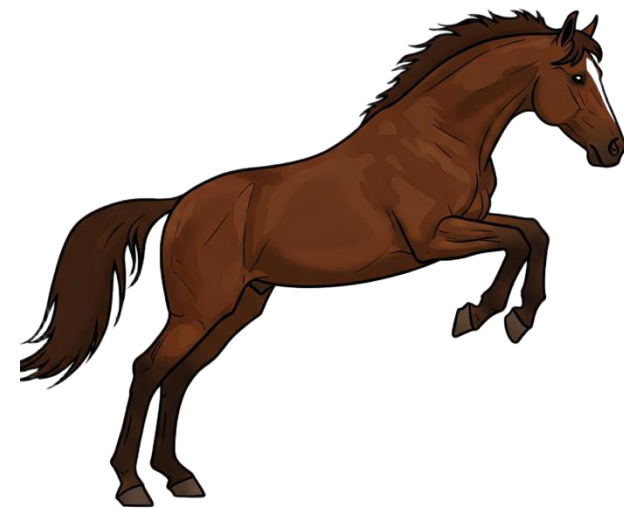


THE TWO L / V RATIOS



HOW DOES THIS HAPPEN?

APPLIED L/V $>$ THRESHOLD L/V



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APPLIED L/V

HOW APPLIED L/V CAN INCREASE?

BIG

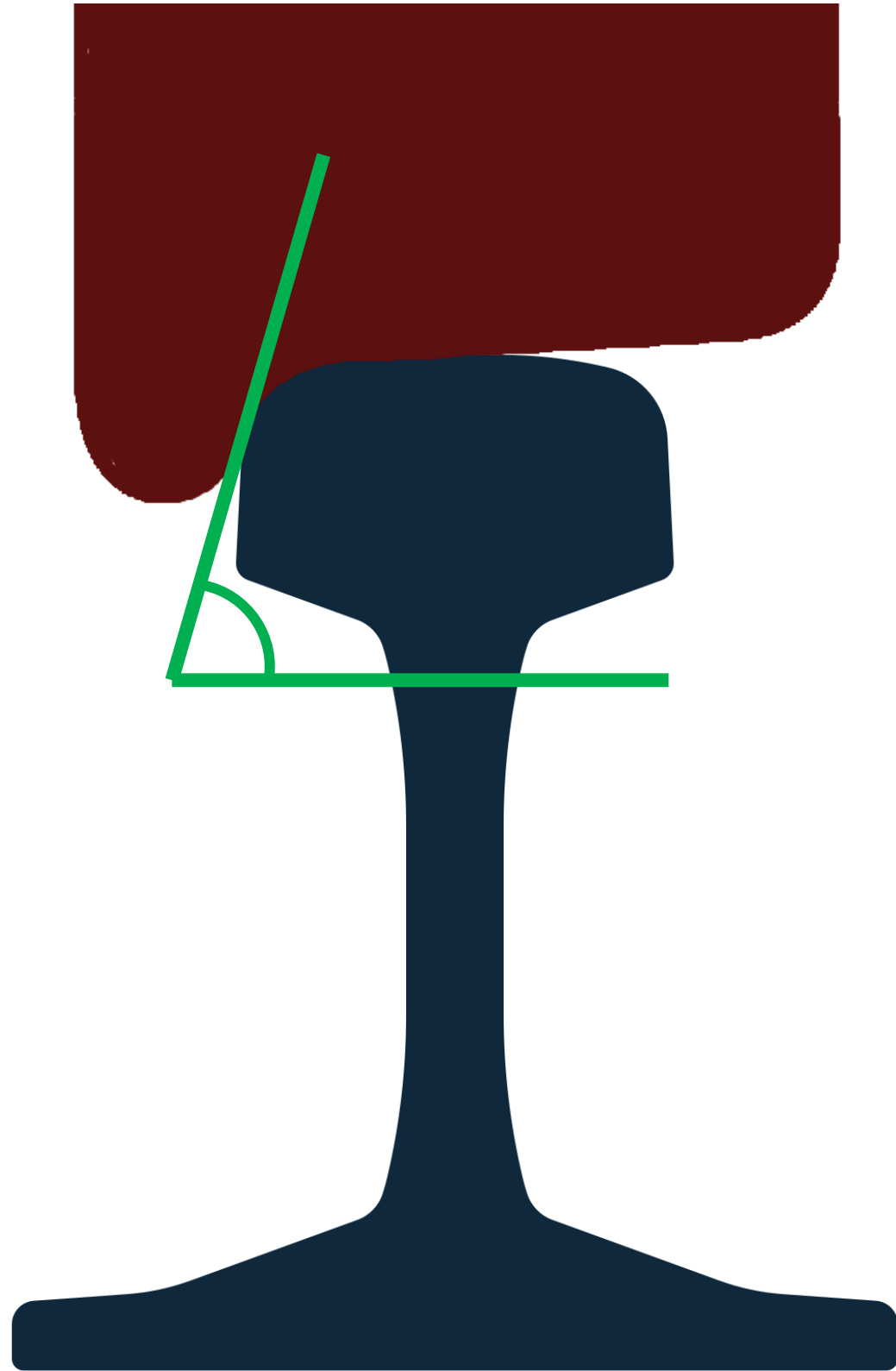
$\frac{L}{V}$

LITTLE

V

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L/V THRESHOLD

WHEEL CLIMB

The threshold that, if exceeded, will result in a **wheel-climb** or rail-rollover for a specific wheel-rail combination.

L/V Threshold for WHEEL CLIMB is a function of:

- Angle at the interface
- Friction at the interface





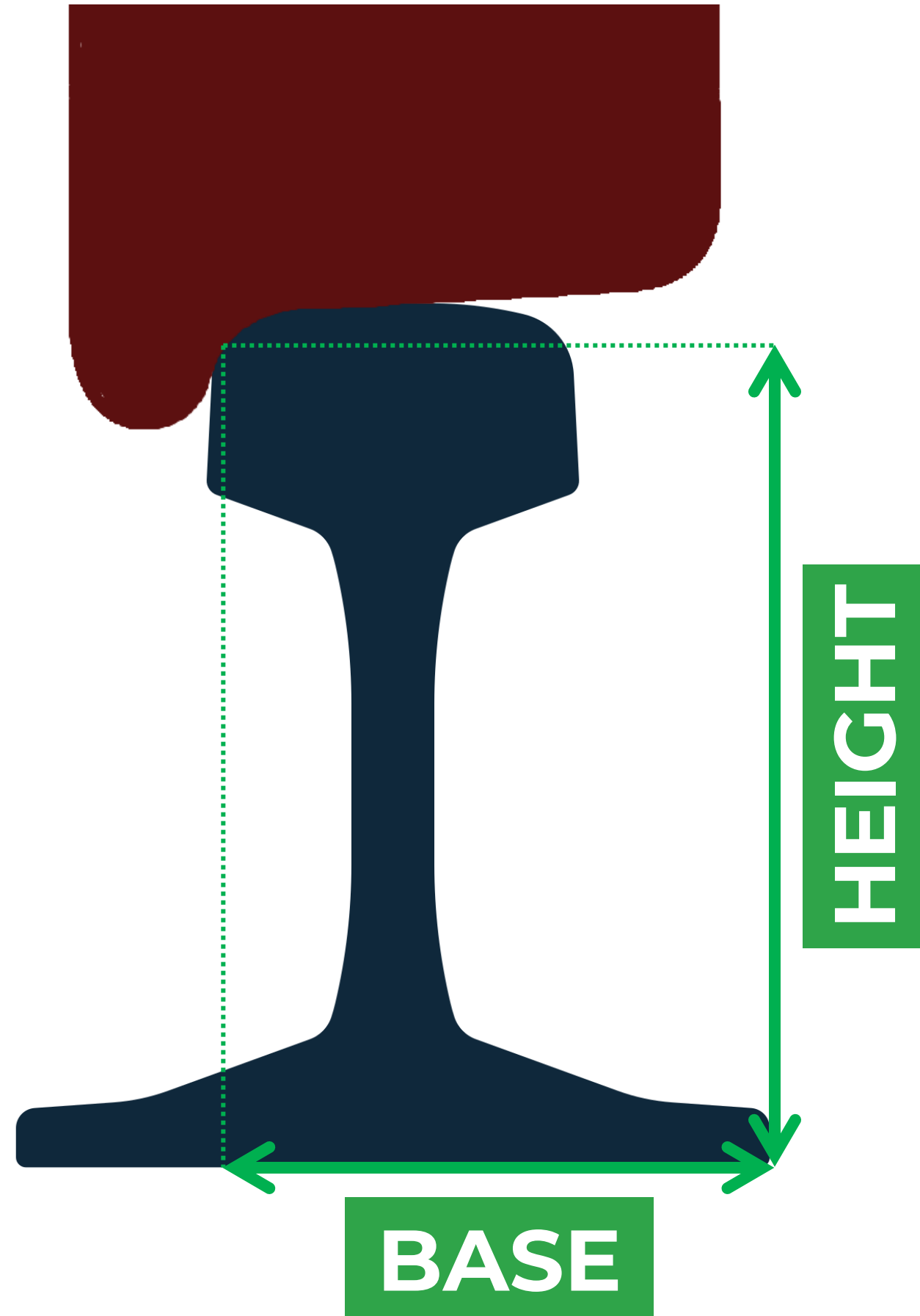
L/V THRESHOLD

WHEEL CLIMB

**MOST WHEEL CLIMB
DERAILMENTS ARE LITTLE V**

APPLIED $\frac{L}{V} = \frac{\text{LATERAL}}{\text{VERTICAL}}$





L/V THRESHOLD

RAIL ROLLOVER

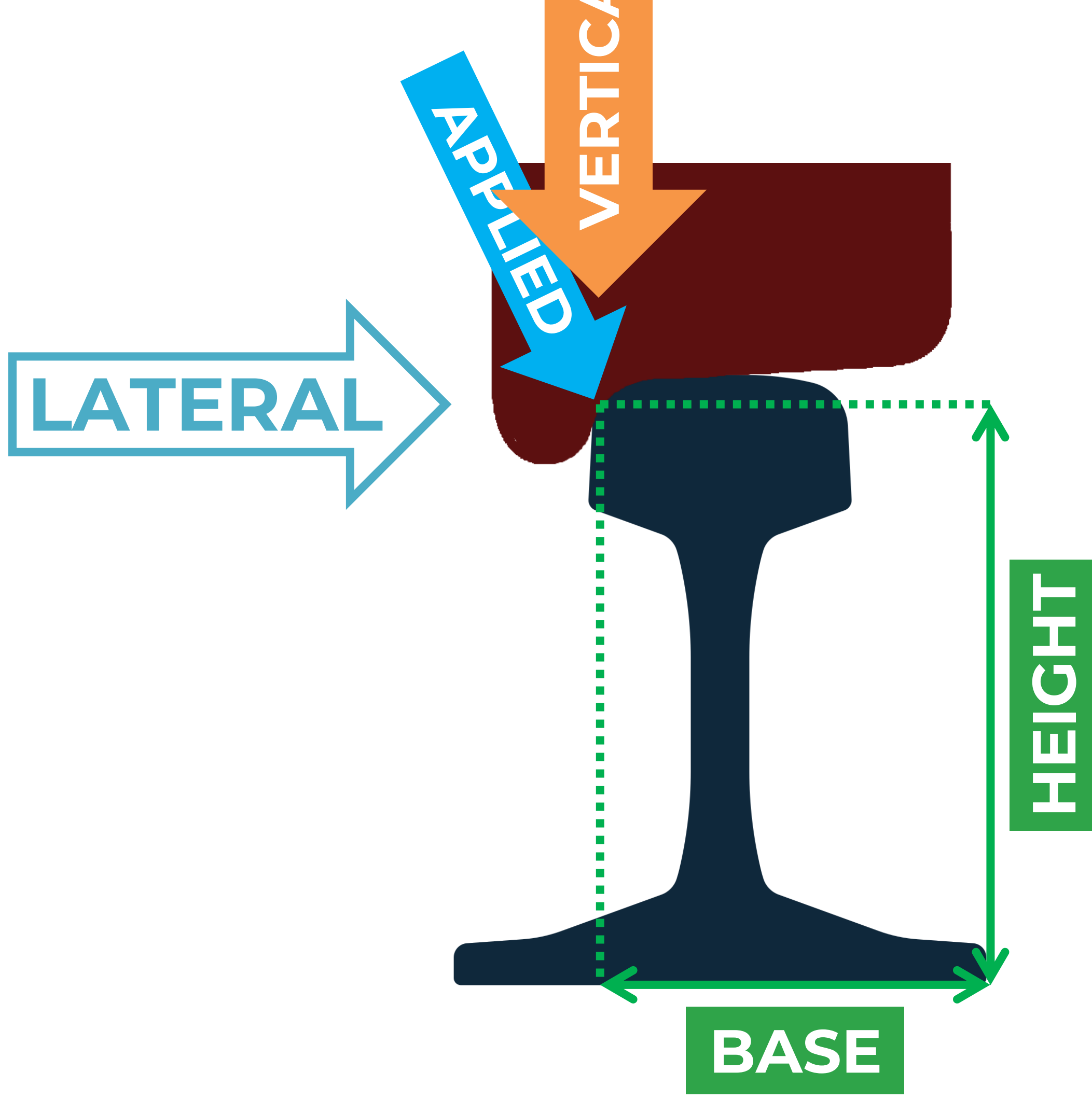
The threshold that, if exceeded, will result in a wheel-climb or **rail-rollover** for a specific wheel-rail combination.

L/V threshold for RAIL-ROLLOVER is a function of:

- The wheel-rail contact point(s)
- The “B/H Ratio”

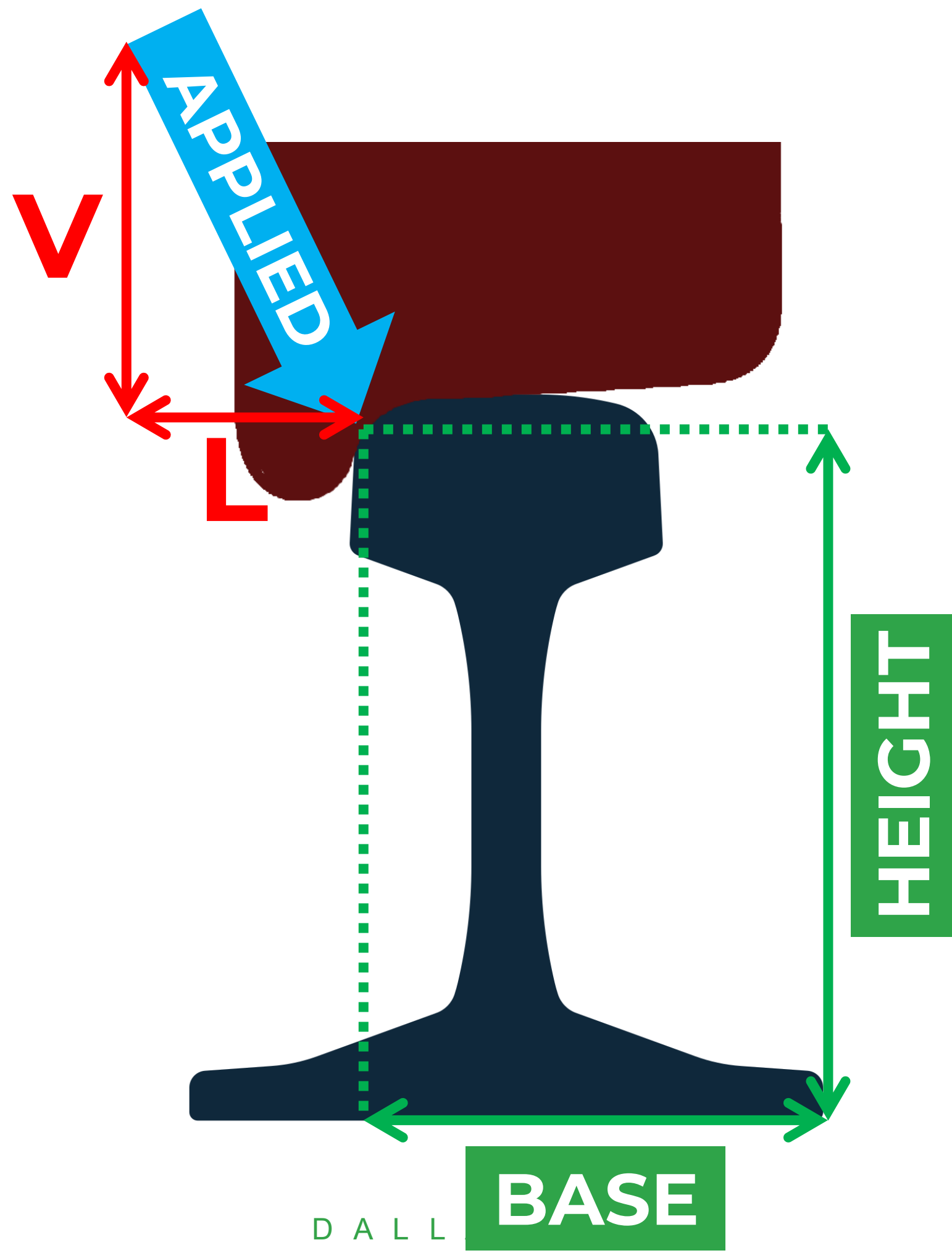
L/V THRESHOLD

RAIL ROLLOVER



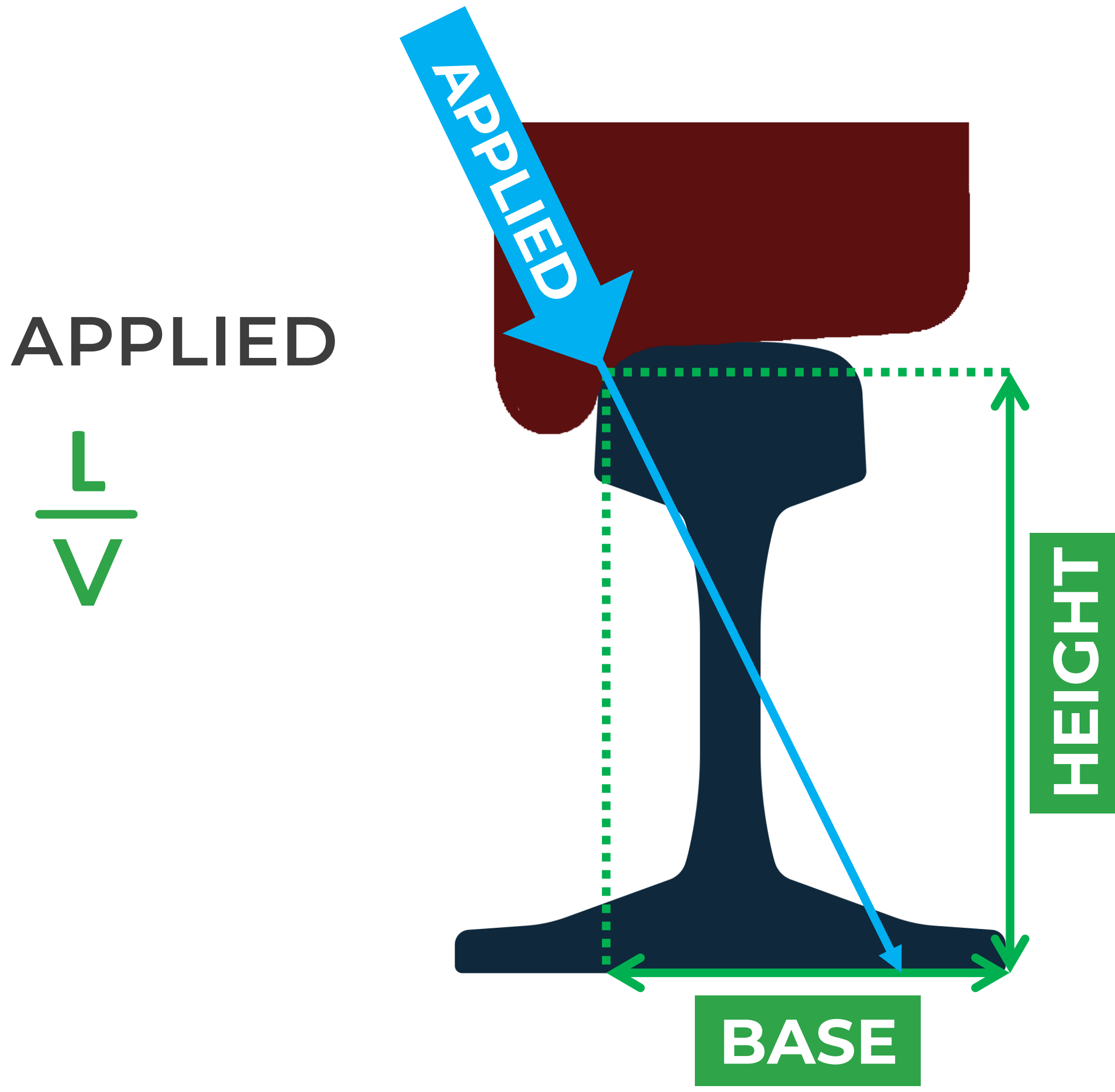
L/V THRESHOLD

RAIL ROLLOVER



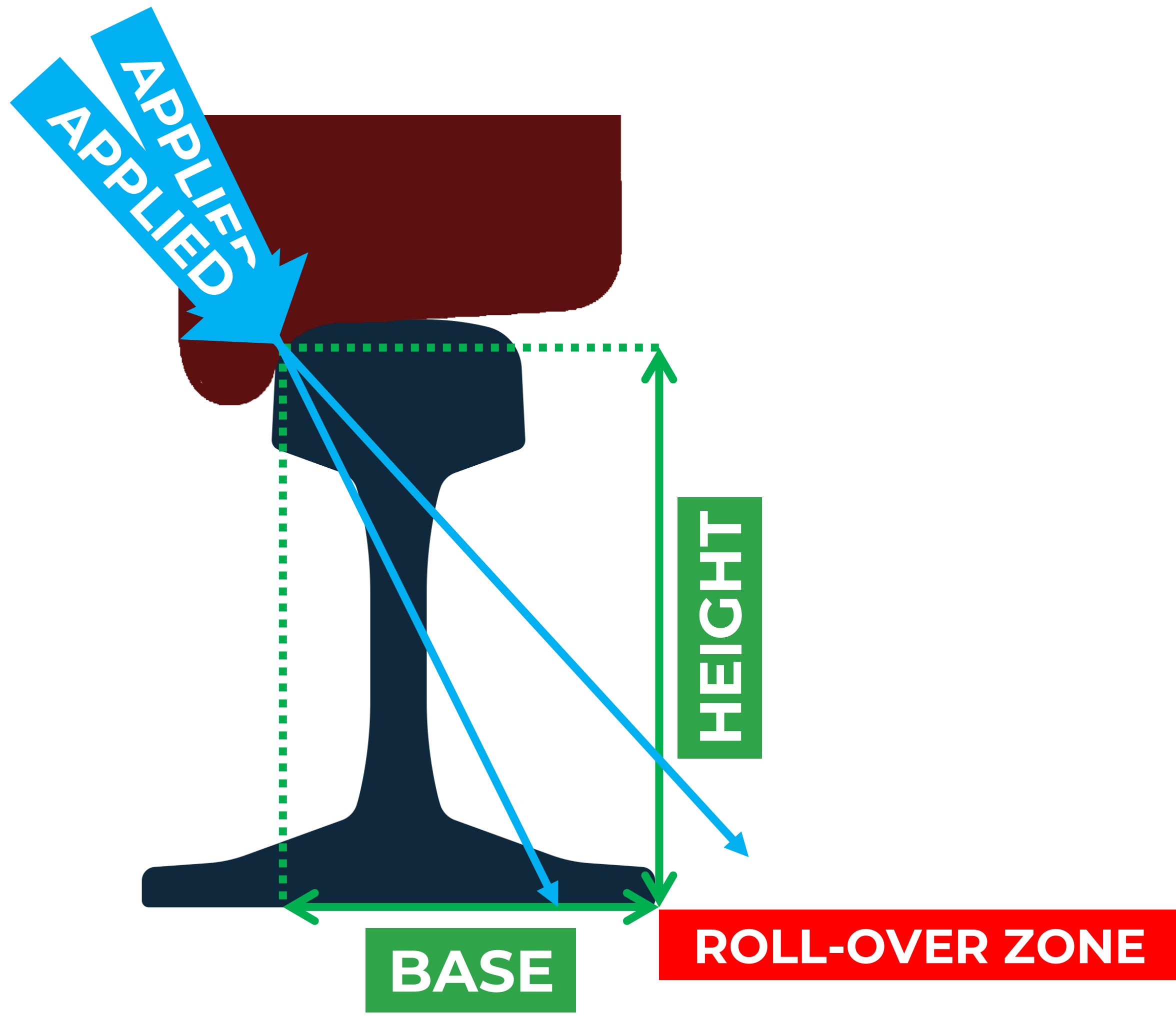
L/V THRESHOLD

RAIL ROLLOVER



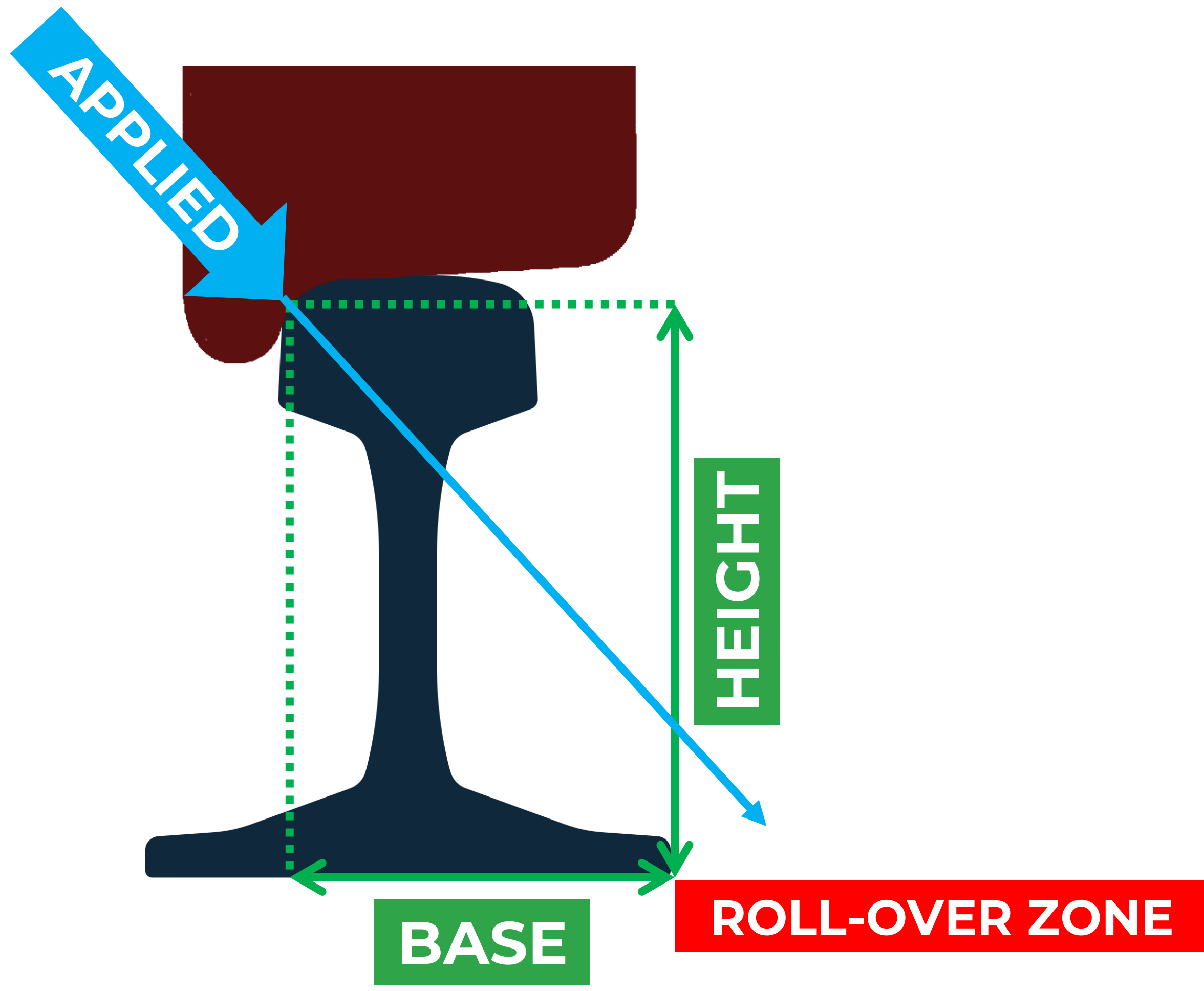
L/V THRESHOLD

RAIL ROLLOVER



L/V THRESHOLD

RAIL ROLLOVER



L/V THRESHOLD RAIL ROLLOVER

**MOST RAIL ROLLOVER
DERAILMENTS ARE BIG L**

$$\frac{L}{V} = \frac{\text{LATERAL}}{\text{VERTICAL}}$$

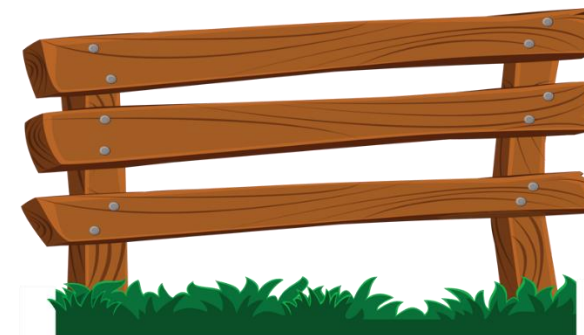
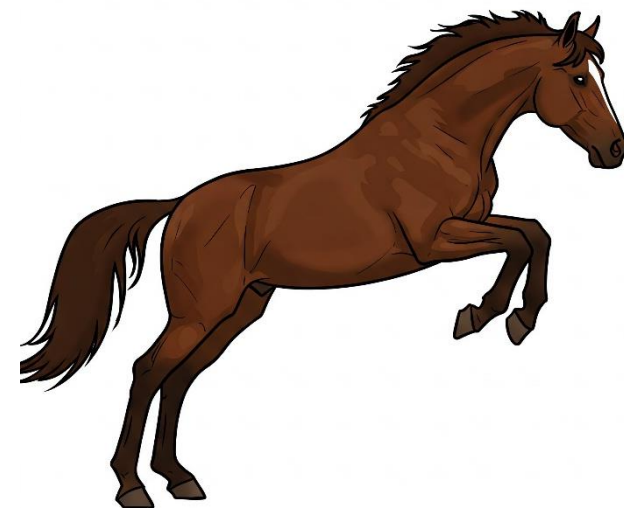
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THE CASE STUDY

- Your phone rings on a Tuesday morning in March.
- A westbound double-stack train derailed overnight in rural North Texas.
- The crew reports that a couple wells are off the track mid-train.



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